Token-Efficient Multimodal Prototyping for Interpretable Cancer Survival Prediction

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Abstract

Accurate cancer survival prediction is critical for personalised treatment planning. Modern approaches increasingly rely on machine learning to integrate high-dimensional histological and genomic data, enabling precise prognostic modelling. While early fusion strategies using transformers have shown promise, existing frameworks struggle to process the vast number of whole-slide image (WSI) patches ($>10^4$) and gene expression profiles (20,000–30,000) without sacrificing intra-modality interactions or computational precision.

In this study, we reproduce the multimodal prototyping (MMP) framework, which aggregates data into concept-based prototypes to reduce complexity. Specifically, WSI patches are compressed into 16 morphological tokens, and gene expressions summarised into 50 pathway-level tokens. We evaluate MMP with transformer-based and optimal transport fusion across four TCGA cancer cohorts, achieving C-indices up to 0.685 and 0.679, respectively, both demonstrating strong predictive performance.

Additionally, we extend MMP by (1) demonstrating that large WSI patch sizes degrade predictive accuracy, reducing the C-index by 3.1%, and (2) incorporating DNA mutation data as a third modality, further improving the C-index to 0.694. Finally, the bidirectional interactions between morphological, transcriptomic, and mutation features, as interpreted from transformer attention weights, provide insights that are valuable for clinical decision-making and for understanding cancer biology.

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1 Introduction

1.1 Cancer Survival Prediction: Clinical Significance

In the UK, approximately 500 people die from cancer every day [3]. As a complex genetic disease with often uncontrollable progression, cancer has become one of the most pressing global health concerns. Its effective treatment increasingly demands targeted and personalised approaches.

The first and most important question for many patients is: *"How long will I live?"* Accurate cancer prognosis—particularly reliable survival analysis—is essential for both patients and clinicians. Cancer survival prediction refers to estimating the probability that a patient will survive beyond a specified time point after diagnosis, and is fundamentally a form of patient risk assessment [4]. Such prognostic information plays a pivotal role in clinical decision-making: it enables more precise patient stratification, informs the selection and intensity of treatment strategies, and helps patients and their families prepare emotionally and financially in anticipation of future outcomes [5].

1.2 Existing Approaches

Standard approaches to survival prediction can be broadly classified into three categories: manual prediction, statistical prediction, and machine learning prediction [6].

Expert-based manual prediction relies heavily on clinicians' experience and subjective assessments of a patient's physical functioning and overall condition. Studies has shown that these methods often produce inconsistent and overly optimistic estimates [7]. Statistical approaches, by contrast, use established models such as Kaplan-Meier[8] and Cox model[9] to quantify survival probabilities based on prognostic factors. These methods improve objectivity and generalisability, but they are limited to low-dimensional data and typically overlook the complex individual differences in disease progression. To address these limitations, and driven by digitalisation in healthcare data, machine learning have emerged as a promising alternative. With the capacity to model complex non-linear relationships, machine learning algorithms can capture underlying patterns in high-dimensional data and generate more personalised risk predictions[4]. These models typically frame survival prediction as one of two tasks: a classification task—grouping patients into risk categories and estimating time-to-event accordingly [10, 11]; or a ranking task—ordering patients by survival risk [1, 12]. For both types of tasks, a variety of machine learning models and frameworks can be applied, with the appropriate selection primarily driven by the data characteristics [4].

1.3 Key Data Modalities

- **Histological Data**

  Pathological assessment remains the gold standard for definitive cancer diagnosis and evaluation of the patient's condition[13]. One of the profound hallmarks of cancer is abnormal cell proliferation, which results in the formation of a tumour [14]. Traditionally, pathology involves trained specialists visually examining cell morphology on tissue slides under a microscope. These phenotypic features reflect underlying genetic alterations within the tumour microenvironment. The introduction of whole-slide imaging (WSI) in 1999 enabled the digitisation of entire tissue slides at high resolution and laid the foundation for computational pathology [13, 15]. Computer-aided analysis of WSIs has helped address the limitations of manual pathology, by reducing the subjectivity and time burden associated with the human interpretation of large-scale histopathological images [16].

Due to the extremely high resolution of whole-slide images (WSIs), which can exceed 10^{10} pixels, a single WSI is typically divided into tens of thousands of small patches. Conventional methods require costly patch-level annotations for supervised classification [13]. In contrast, multiple instance learning (MIL) has now become widely adopted for WSI modelling as it offers a weakly supervised alternative that relies only on slide-level labels [13, 17, 18, 19, 20]. MIL models extract patch-level embeddings and pool them into slide-level representations for downstream tasks. This approach has demonstrated strong performance in applications ranging from tumour detection[18, 19] to survival prediction [1, 20].

- **Genomic Data**

  Cancer is a genetic disease characterised by DNA mutations and abnormal gene expression. Different types of genomic data can be used for predictive modelling, including mRNA expression, somatic mutations, DNA methylation, and copy number alterations [21]. This study focuses on mRNA expression and somatic mutations, which are complementary sources of information. Mutations reflect underlying genetic alterations that may initiate cancer, while expression measurements reflect the downstream cell behaviour and the activity of the tumour.

  Gene expression information is captured by mRNA gene counts, now commonly obtained through RNA sequencing and provided in tabular formats containing expression values. These data offer valuable molecular profiles reflecting the differential activation of genes involved in cancer. The main challenge in using these data is the intrinsic curse of dimensionality: the number of samples (typically 300–1,000) is much smaller than the number of measured genes (often 10,000–60,000) [22]. This imbalance can lead to overfitting and poor generalisation. A common strategy to address this is to incorporate biological prior knowledge and filter for relevant gene signatures and biomarkers—i.e., one or a group of genes associated with a certain prediction target [21, 22, 23, 24]. Alternatively, dimensionality reduction techniques such as Principal Component Analysis (PCA) can be used to extract compact representations [21, 22, 23, 25].

  Genetic mutation data, commonly obtained from Illumina DNA sequencing, are provided in tabular formats with binary indicators of mutation events. These data include key drivers of cancer progression, but remain underutilised due to their high dimensionality and sparsity, as most genes do not exhibit mutations in a given sample [26]. Similar to gene expression preprocessing, existing studies either integrate mutation information with biological pathway annotations [26, 27, 28] or derive low-dimensional representations through feature extraction [26, 29].

1.4 Multimodal Framework

In modern clinical practice, pathologists integrate histological and genomic data to improve patient prognostication, as these modalities provide complementary information on tumour morphology and molecular alterations, respectively [30]. To fully leverage this rich heterogeneous data, framing cancer survival prediction as a multimodal learning task in machine learning has emerged as an effective strategy [31]. This approach has demonstrated improved predictive performance over single-modality methods [32, 33].

The main barrier in multimodal frameworks is determining robust strategies to fuse the heterogeneous modalities of histological and genomic data. Many previous studies have employed late fusion techniques, in which each modality is transformed into its corresponding high-level feature embedding by separate encoders. These embeddings are then merged to form a joint patient representation for the final prediction [34]. A major limitation is that this architecture cannot capture interactions between modalities and therefore loses important interpretability across morphological and molecular features [35, 36, 37].

Early fusion enables modelling of multimodal interactions. In contrast to late fusion, it treats WSI patch embeddings and gene groups as tokens, combining them into a joint input sequence. The joint tokens are then processed by a single encoder to generate integrated patient representations for prediction [34]. This encoder employs a Transformer to capture cross-modal interactions through self-attention over tokens from all modalities. Transformer fusion is a common early fusion technique, in which the joint input sequence is obtained by directly concatenating the modality-specific tokens[38, 39, 40]. One alternative is optimal transport (OT) cross-modal alignment [24]. Unlike Transformer fusion, OT fusion combines modality-specific tokens by learning an optimal transport plan to align them through minimising their overall matching cost.

Previous work has shown that early fusion strategies can outperform late fusion in downstream cancer prediction tasks [41]. However, inputting a large number of tokens—primarily from histology patches—leads to substantial computational complexity and cost. Prior approaches mitigate this computational burden by using attention approximation [42], cross-attention [41], or batch approximation [24]. Nevertheless, these strategies inherently suffer from approximation errors and information loss. Moreover, they still need to process large numbers of WSI tokens without fundamentally reducing their redundancy [43].

Recently, Song et al. proposed a framework called MultiModal Prototyping (MMP) to resolve the above limitations [1]. MMP enables token-efficient multimodal cancer survival prediction with strong bidirectional interpretability. This prototype-based approach can accommodate both Transformer fusion and OT cross-modal alignment techniques without relying on computational approximations. Its core idea is to use a small set of meaningful prototypes to replace the original dense tokens in each modality before multimodal fusion. Their work implements two data modalities—histology and gene expression information—to perform this prediction task. Specifically, WSI patch embeddings are aggregated into 16 key morphological prototypes per slide through unsupervised learning. Genomic information is filtered and mapped into 50 Cancer Hallmark pathway prototypes. MMP demonstrated outstanding performance with significantly fewer operations under both Transformer and OT fusion across six cancer cohorts. It also provided concept-based insights into the connections between histopathology and genomics. We refer to this original work as *the original MMP study* in the following sections.

1.5 Aims and Objectives

This dissertation has two main objectives: first, to reproduce and validate the MMP framework described in *the original MMP study* by (1) applying it to survival prediction on four out of the original six cancer cohorts selected from The Cancer Genome Atlas (TCGA), and (2) visualising and interpreting the learned cross-modal interactions; second, to extend and enhance the framework by (3) assessing the impact of WSI patch size on predictive performance, and (4) incorporating a third data modality—DNA mutation—to improve predictive accuracy.

2 Methodology

We followed the procedure described in *the original MMP study* to build the multimodal framework. Due to the extremely time-consuming nature of WSI patch embedding extraction, the *preprocessing* described in Section 2.2.1 utilises readily available patch embeddings generated with different settings than those in *the original MMP study*. All other reproduced components strictly adhere to the original configuration. Extensions beyond *the original MMP study* are explicitly indicated by the "**EXTENSION**" labels in the section headings.

In this work, the following notation conventions are adopted: z denotes a scalar, $\mathbf{z}$ indicates a vector, and $\mathbf{Z}$ denotes a matrix. For a collection of vectors $\{\mathbf{z}_c\}_{c=1}^{C} \in \mathbb{R}^d$, the matrix $\mathbf{Z} \in \mathbb{R}^{C\times d}$ is defined such that each row corresponds to $\mathbf{z}_c$. The concatenation of matrices is expressed by $[\mathbf{X}, \mathbf{Z}]$.

2.1 Prototype-based Token Construction

2.1.1 Histology Tokens

Our goal is to summarise the tissue information that follows a limited set of recurring morphological patterns into key prototypes with distinct biological meanings. For example, separate prototypes may represent tumour cells, connective tissue, immune cells, and other structures. This process is performed in an unsupervised manner.

Preprocessing. Each whole-slide image (WSI) is tessellated into N_h non-overlapping patches of 1024×1024 pixels at 40× magnification (0.25 μm/pixel). Each patch is subsequently encoded into an embedding using a pretrained encoder f_{enc}, such that $\mathbf{z}_{i,h} = f_{\text{enc}}(\mathbf{x}_{i,h}) \in \mathbb{R}^D$. We denote the set of patch embeddings for a slide as $\mathbf{S}_h = \{\mathbf{z}_{i,h}\}_{i=1}^{N_h}$.

It should be noted that *the original MMP study* used 256×256 pixel patches extracted at 20× magnification (0.5 μm/pixel).

Prototyping. K-means clustering is applied to *all* patch embeddings *in the training set* to obtain C_h morphological prototypes, denoted as $\{\mathbf{a}_{c,h}\}_{c=1}^{C_h}$, where each $\mathbf{a}_{c,h} \in \mathbb{R}^D$. These K-means prototypes are used as initial cluster centers for subsequent aggregation applied to patch embeddings within each WSI.

Histology Aggregation. For each slide, we aggregate the patch embeddings $\mathbf{S}_h$ guided by the prototypes $\{\mathbf{a}_{c,h}\}_{c=1}^{C_h}$ into a set of slide tokens $\mathbf{S}_{\text{slide}} \in \mathbb{R}^{C_h\times d_h}$. This is performed via Gaussian Mixture Models (GMM). In GMM, the probability distribution of each patch embedding $\mathbf{z}_{i,h}$ is defined as

$$\begin{aligned} p(\mathbf{z}_{i,h};\theta) &= \sum_{c=1}^{C_h} p(c_i = c)\,\cdot\, p(\mathbf{z}_{i,h} \mid c_i = c) \\ &= \sum_{c=1}^{C_h} \pi_c \,\cdot\, \mathcal{N}\big(\mathbf{z}_{i,h};\, \boldsymbol{\mu}_c,\, \boldsymbol{\Sigma}_c\big). \end{aligned}$$

where c_i denotes the mixture identity of $\mathbf{z}_{i,h}$, and $\theta = \{\pi_c, \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c\}_{c=1}^{C_h}$ denotes the mixture probability, means, and diagonal covariances. Within this framework, $\boldsymbol{\mu}_c$ captures a representative morphological pattern present in the slide, while π_c reflects the proportion of regions exhibiting this pattern. We initialise $\boldsymbol{\mu}_c$ using the K-means prototypes $\mathbf{a}_{c,h}$ and $\boldsymbol{\Sigma}_c$ as the identity matrix. These parameters are iteratively updated to maximise the likelihood:

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{N_h} \log p(\mathbf{z}_{i,h};\theta),$$

via an expectation-maximization (EM) procedure implemented as a feedforward network operation. Each post-aggregation slide token is defined as

$$\mathbf{z}_{c,h}^{\text{agg.}} = \left[\hat{\pi}_c, \hat{\boldsymbol{\mu}}_c, \hat{\boldsymbol{\Sigma}}_c\right] \in \mathbb{R}^{d_h},$$

and the complete slide representation is given by

$$\mathbf{S}_{\text{slide}} = \left\{\mathbf{z}_{c,h}^{\text{agg.}}\right\}_{c=1}^{C_h},$$

which provides a comprehensive description of the WSI.

The use of soft clustering in this aggregation quantifies the contributions from each patch embedding $\mathbf{z}_{i,h}$ to all morphological prototypes, thereby enabling more fine-grained preservation of information.

Impact on Input Size. The histology prototyping process reduces the initial set of N_h patch embeddings—which is large and variable across slides (typically $N_h > 10^4$ per WSI)—into a fixed collection of C_h slide tokens. In this work, we set $C_h = 16$, resulting in over a 600-fold reduction in the input size for the histology modality in the multimodal fusion model.

2.1.2 Genomics Tokens

We aim to summarise omic profiles into pathway prototypes. Specifically, the redundant gene list is reduced to concise tokens that encode interpretable biological signals. To this end, we employ the Hallmark Gene Sets as prototypes, each representing a defined cellular process with coherent expression patterns, such as DNA repair, protein secretion, and others, that are relevant to cancer survival [44].

Gene Expression Prototyping. For each patient, the genomic data are represented as $\mathbf{x}_g = \{x_{i,g}\}_{i=1}^{N_g}$, where N_g denotes the total number of genes in the measurement. We define C_g pathway prototypes (C_g = 50) derived from the existing Hallmark Gene Sets, denoted by $\{\mathbf{a}_{c,g}\}_{c=1}^{C_g}$, where each binary vector $\mathbf{a}_{c,g} \in \{0,1\}^{N_g}$ indicates the inclusion of genes in pathway c (1 if present, 0 otherwise).

Gene Expression Aggregation. In this section, the additional subscript "ex" is used to explicitly indicate gene expression data. Each gene measurement in $\mathbf{x}_{g,ex} = \{x_{i,g,ex}\}_{i=1}^{N_g}$ satisfies $x_{i,g,ex} \in \mathbb{R}$. For each patient, we aggregate the gene expressions, guided by the prototypes $\{\mathbf{a}_{c,g}\}_{c=1}^{C_g}$, into a set of pathway tokens $\mathcal{S}_{\text{expression}} \in \mathbb{R}^{C_g \times N_{c,g}}$, where $N_{c,g}$ denotes the number of genes included in pathway c. Each aggregated gene expression token for pathway c is defined as

$$\mathbf{z}_{c,g,ex}^{\text{agg.}} = R\left(\mathbf{x}_{g,ex} \odot \mathbf{a}_{c,g}\right) \in \mathbb{R}^{N_{c,g}},$$

where $\odot$ denotes element-wise multiplication that retains only genes included in pathway c, and R removes zeros to produce a compact representation. The complete set of aggregated pathway tokens is given by

$$\mathcal{S}_{\text{expression}} = \left\{\mathbf{z}_{c,g,ex}^{\text{agg.}}\right\}_{c=1}^{C_g},$$

providing a concise representation of the gene expression profile. Note that the length of each pathway token $N_{c,g}$ varies depending on the number of genes defined for each pathway.

EXTENSION: Mutation Prototyping. We extend the original framework to a third data modality: DNA mutations. To summarise the sparse, high-dimensional mutation data into compact pathway representations, we employed the same Hallmark prototypes defined for gene expression analysis. This consistent approach enables pathway-level interpretability across the data modalities and its validity is supported by the following evidence.

Figure 1 shows their overlap with mutated genes across the cancer cohorts included in this study, and Table 1 highlights which frequently mutated genes are retained or excluded. The consistently high overlap indicates that many mutations occur within well-characterised biological processes. Notably, genes such as *TTN* and *MUC16*, which are among the largest in the human genome and frequently accumulate incidental mutations, are excluded. In contrast, recurrently mutated drivers like *TP53* and *VHL* are retained. This prototyping removes uninformative signals while preserving established cancer-relevant alterations.

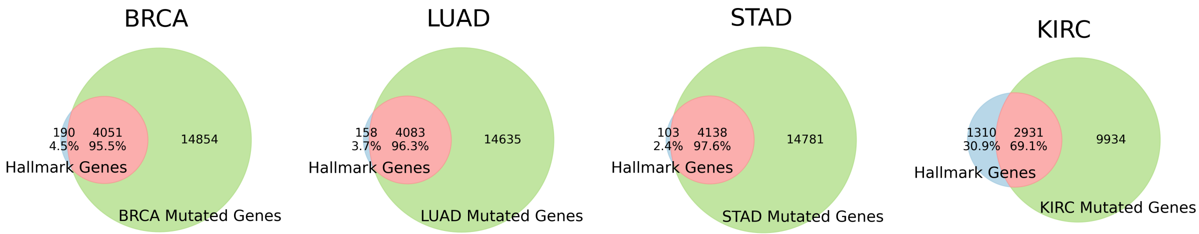

Figure 1: Venn diagrams showing the overlap between Hallmark genes and mutated genes across four cancer types. Each subplot reports the absolute counts and the proportion of Hallmark genes covered by the observed mutations.

Table 1: The top three most frequently mutated genes identified in each cancer type. Genes included in the Hallmark set are indicated in underlined bold. Further details of the studied cancers are provided in Section 3.1.

Rank	BRCA	LUAD	STAD	KIRC
1st	TTN	TTN	TTN	<u>**VHL**</u>
2nd	<u>**PIK3CA**</u>	MUC16	MUC16	PBRM1
3rd	<u>**TP53**</u>	<u>**RYR2**</u>	<u>**SYNE1**</u>	TTN

EXTENSION: Mutation Aggregation. In this section, the subscript "mt" denotes mutation data. Each gene measurement is represented as $\mathbf{x}_{g,mt} = \{x_{i,g,mt}\}_{i=1}^{N_g}$, with $x_{i,g,mt}$ defined by one of the following strategies, the impact of which is compared in downstream tasks.

- **Binary encoding**: $x_{i,g,mt} \in \{0, 1\}$, set to 1 if any mutation is observed in gene i, 0 otherwise.
- **Weighted encoding**: $x_{i,g,mt} \in \mathbb{R}_{\geq 0}$, computed as the sum of weights, assigned according to the TCGA-annotated Variant Effect Predictor (VEP) impacts [2], of all mutations observed in gene i (Table 2).

Similar to gene expression aggregation, for each patient, we aggregate the mutation measurements guided by the prototypes $\{\mathbf{a}_{c,g}\}_{c=1}^{C_g}$ into a set of pathway tokens $\mathcal{S}_{\text{mutation}} \in \mathbb{R}^{C_g \times N_{c,g}}$. Each mutation token for pathway c is defined as

$$\mathbf{z}_{c,g,mt}^{\text{agg.}} = \pi\left(\mathbf{x}_{g,mt}\right) \in \mathbb{R}^{N_{c,g}},$$

where π denotes the selection operator that selects the entries corresponding to genes included in pathway c. The complete set of aggregated mutation pathway tokens is given by

$$\mathcal{S}_{\text{mutation}} = \left\{\mathbf{z}_{c,g,mt}^{\text{agg.}}\right\}_{c=1}^{C_g},$$

providing a compact representation of the DNA mutation profile.

Table 2: Mapping of TCGA-annotated VEP mutation impacts to quantitative weights [2].

Impact Annotation	VEP Description	Weight
HIGH	Disruptive, likely to cause loss of protein function	5.0
MODERATE	Non-disruptive, may alter protein effectiveness	3.0
LOW	Mostly harmless, unlikely to change protein function	1.0
MODIFIER	Non-coding variants with no predicted protein impact	0.2
NO MUTATION	No mutation detected	0

Impact on Input Size. For both gene expression and mutation, pathway prototyping reduces the original large profile per patient ($N_g \simeq 2\text{–}3 \times 10^4$) to a smaller set of hallmark genes ($N_{\text{hallmark}} \simeq 4 \times 10^3$) distributed across 50 pathways (each with $N_{c,g} < 200$). This achieves approximately a 5-10$\times$ reduction in data size while providing concept-based interpretability.

In summary, we obtain three sets of prototype-based representations: the histology slide tokens $\mathbf{S}_{\text{slide}} \in \mathbb{R}^{C_h \times d_h}$, the RNA expression tokens $\mathbf{S}_{\text{expression}} \in \mathbb{R}^{C_g \times N_{c,g}}$, and the DNA mutation tokens $\mathbf{S}_{\text{mutation}} \in \mathbb{R}^{C_g \times N_{c,g}}$. Each set encodes complementary aspects of the patient profile and together they enable token-efficient multimodal fusion.

2.2 Multimodal fusion

To integrate histology and genomic tokens with differing dimensions, we first perform dimensionality matching.

For histology representations, a linear projection

$$\mathbf{z}_{c,h}^{\text{pre}} = f_h^{\text{pre}}\left(\mathbf{z}_{c,h}^{\text{agg}}\right) \in \mathbb{R}^d$$

is applied to map the input to the common embedding space of dimension d.

For each pathway, self-normalising neural networks (SNNs) are employed to map each variable-length feature to fixed-length embeddings [45]:

$$\mathbf{z}_{c,g}^{\text{pre}} = f_{c,g}^{\text{pre}}\left(\mathbf{z}_{c,g}^{\text{agg}}\right) \in \mathbb{R}^d.$$

All projection parameters f_h^{pre} and $\{f_{c,g}^{\text{pre}}\}$ are jointly optimised during downstream training.

A distinctive component of the MMP framework is the inclusion of prototype encodings to enrich the embeddings with consistent prototype identity information [46]. For each prototype c, we generate a learnable embedding $\mathbf{e}_c \in \mathbb{R}^{d_e}$ with random initialisation ($d_e = 32$). The embeddings are then concatenated as follows:

$$\mathbf{z}_{c,h}^{\text{pre}} = \left[\mathbf{z}_{c,h}^{\text{pre}}, \mathbf{e}_c\right], \quad \mathbf{z}_{c,g,\text{ex}}^{\text{pre}} = \left[\mathbf{z}_{c,g,\text{ex}}^{\text{pre}}, \mathbf{e}_c\right], \quad \mathbf{z}_{c,g,\text{mt}}^{\text{pre}} = \left[\mathbf{z}_{c,g,\text{mt}}^{\text{pre}}, \mathbf{e}_c\right].$$

As an extension, the incorporation of DNA mutation as a third modality is implemented exclusively within the Transformer-based fusion module.

2.2.1 Transformer Attention

Bi-modal Attention. Following the original MMP formulation, we first consider bi-modal fusion of histology and gene expression tokens. The query, key, and value matrices are defined as:

$$\mathbf{Q} = \begin{pmatrix} \mathbf{Q}_{g,ex} \\ \mathbf{Q}_h \end{pmatrix} = \begin{pmatrix} \mathbf{Z}_{g,ex}^{\text{pre}} \\ \mathbf{Z}_h^{\text{pre}} \end{pmatrix} \mathbf{W}_Q, \quad \mathbf{K} = \begin{pmatrix} \mathbf{K}_{g,ex} \\ \mathbf{K}_h \end{pmatrix} = \begin{pmatrix} \mathbf{Z}_{g,ex}^{\text{pre}} \\ \mathbf{Z}_h^{\text{pre}} \end{pmatrix} \mathbf{W}_K, \quad \mathbf{V} = \begin{pmatrix} \mathbf{V}_{g,ex} \\ \mathbf{V}_h \end{pmatrix} = \begin{pmatrix} \mathbf{Z}_{g,ex}^{\text{pre}} \\ \mathbf{Z}_h^{\text{pre}} \end{pmatrix} \mathbf{W}_V,$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d}$ are learnable projection matrices.

The standard Transformer attention is then computed as [47]:

$$\mathbf{Z}^{\text{post}}_{g,ex+h} = \sigma\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V} \quad \in \mathbb{R}^{(C_{g,ex}+C_h)\times d},$$

where $\sigma(\cdot)$ denotes row-wise softmax.

This can be further decomposed into intra-modal and cross-modal interactions:

$$\mathbf{Z}^{\text{post}}_{g,ex+h} = \sigma\left(\frac{1}{\sqrt{d}}\begin{pmatrix}\mathbf{Q}_{g,ex}\mathbf{K}^\top_{g,ex} & \mathbf{Q}_{g,ex}\mathbf{K}^\top_{h} \\ \mathbf{Q}_{h}\mathbf{K}^\top_{g,ex} & \mathbf{Q}_{h}\mathbf{K}^\top_{h}\end{pmatrix}\right)\begin{pmatrix}\mathbf{V}_{g,ex} \\ \mathbf{V}_h\end{pmatrix}.$$

This enables both intra-modal self-attention ($g, ex \to g, ex$, $h \to h$) and bidirectional cross-modal attention ($g, ex \to h$, $h \to g, ex$).

EXTENSION: Tri-modal Attention. To incorporate a third modality (e.g., DNA mutation pathways), the formulation extends to:

$$\mathbf{Q} = \begin{pmatrix}\mathbf{Q}_{g,ex} \\ \mathbf{Q}_{g,mt} \\ \mathbf{Q}_h\end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix}\mathbf{K}_{g,ex} \\ \mathbf{K}_{g,mt} \\ \mathbf{K}_h\end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix}\mathbf{V}_{g,ex} \\ \mathbf{V}_{g,mt} \\ \mathbf{V}_h\end{pmatrix},$$

and the output is:

$$\mathbf{Z}^{\text{post}}_{g,ex+g,mt+h} = \sigma\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V} \quad \in \mathbb{R}^{(C_{g,ex}+C_{g,mt}+C_h)\times d}.$$

This can be explicitly decomposed as:

$$\sigma\left(\frac{1}{\sqrt{d}}\begin{pmatrix}\mathbf{Q}_{g,ex}\mathbf{K}^\top_{g,ex} & \mathbf{Q}_{g,ex}\mathbf{K}^\top_{g,mt} & \mathbf{Q}_{g,ex}\mathbf{K}^\top_{h} \\ \mathbf{Q}_{g,mt}\mathbf{K}^\top_{g,ex} & \mathbf{Q}_{g,mt}\mathbf{K}^\top_{g,mt} & \mathbf{Q}_{g,mt}\mathbf{K}^\top_{h} \\ \mathbf{Q}_{h}\mathbf{K}^\top_{g,ex} & \mathbf{Q}_{h}\mathbf{K}^\top_{g,mt} & \mathbf{Q}_{h}\mathbf{K}^\top_{h}\end{pmatrix}\right)\begin{pmatrix}\mathbf{V}_{g,ex} \\ \mathbf{V}_{g,mt} \\ \mathbf{V}_h\end{pmatrix}.$$

This formulation allows all pairwise and intra-modal interactions among histology, gene expression, and mutation modalities.

Computation Complexity. For bi-modal attention, the complexity is reduced from the original $O\big((N_{g,ex}+N_h)^2\big)$ to $O\big((C_{g,ex}+C_h)^2\big)$ through prototyping. In the tri-modal setting, the original dense attention over all raw tokens would require $O\big((N_{g,ex}+N_{g,mt}+N_h)^2\big)$, whereas the proposed approach reduces this to $O\big((C_{g,ex}+C_{g,mt}+C_h)^2\big)$, which remains substantially lower and more scalable.

2.2.2 Optimal Transport Cross-alignment

Inspired by unbalanced optimal transport (OT), we model cross-modal interactions as the problem of learning an alignment plan $\mathbf{T} \in \mathbb{R}_+^{C_g\times C_h}$ that minimises the overall pairing cost between modalities. This can be formulated as an entropic-regularised OT problem [48]:

$$\min_{\mathbf{T}} \sum_{c,c'} \mathbf{D}_{c,c'}\,\mathbf{T}_{c,c'} + \epsilon\,\mathbf{T}_{c,c'}\log\mathbf{T}_{c,c'},$$

subject to the marginal constraints that enforce equal contribution of each prototype to the overall transport plan:

$$\sum_{c=1}^{C_g}\mathbf{T}_{c,c'} = \frac{1}{C_h}, \quad \sum_{c'=1}^{C_h}\mathbf{T}_{c,c'} = \frac{1}{C_g}.$$

Here, $\mathbf{D}_{c,c'}$ denotes the pairwise cost between tokens, typically computed using the L_2 distance, and ϵ is a regularisation parameter that ensures smoothness of the solution. The optimal plan $\hat{\mathbf{T}}$ can be efficiently obtained via the Sinkhorn algorithm [49].

This alignment provides strong interpretability as it explicitly encodes the optimal cross-modal correspondence. For example, $\hat{\mathbf{T}}\mathbf{Z}_h^{\text{pre}} \in \mathbb{R}^{C_g \times d}$ represents the projection of histology information into the gene expression prototypes ($h \to g$), while $\hat{\mathbf{T}}^\top \mathbf{Z}_g^{\text{pre}} \in \mathbb{R}^{C_h \times d}$ corresponds to projecting gene expression information into histology space ($g \to h$).

Based on the aligned representations, we then perform intra-modal self-attention. Specifically, we define:

$$\mathbf{Q}_g = (\hat{\mathbf{T}}\mathbf{Z}_h^{\text{pre}})\mathbf{W}_Q, \quad \mathbf{K}_g = (\hat{\mathbf{T}}\mathbf{Z}_h^{\text{pre}})\mathbf{W}_K, \quad \mathbf{V}_g = (\hat{\mathbf{T}}\mathbf{Z}_h^{\text{pre}})\mathbf{W}_V,$$
$$\mathbf{Q}_h = (\hat{\mathbf{T}}^\top\mathbf{Z}_g^{\text{pre}})\mathbf{W}_Q, \quad \mathbf{K}_h = (\hat{\mathbf{T}}^\top\mathbf{Z}_g^{\text{pre}})\mathbf{W}_K, \quad \mathbf{V}_h = (\hat{\mathbf{T}}^\top\mathbf{Z}_g^{\text{pre}})\mathbf{W}_V,$$

where $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are learnable projection matrices. The intra-modal Transformer attention is computed as:

$$\mathbf{Z}_h^{\text{post}} = \sigma\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d}}\right)\mathbf{V}_h, \quad \mathbf{Z}_g^{\text{post}} = \sigma\left(\frac{\mathbf{Q}_g \mathbf{K}_g^\top}{\sqrt{d}}\right)\mathbf{V}_g.$$

Computation Complexity. Compared to dense OT alignment over all raw tokens, whose complexity scales as $O(N_g \times N_h)$ per iteration, the proposed approach reduces this to $O(C_g \times C_h)$ through prototyping, followed by intra-modal self-attention with complexity $O(C_g^2 + C_h^2)$.

2.3 Survival Prediction

The post-attention embeddings are processed by prototype-specific feedforward networks $\{f_c^{\text{post}}\}$, followed by layer normalisation (LN). The resulting embeddings are averaged within each modality and concatenated to form a single patient-level representation:

$$\mathbf{z}_{\text{patient}}^{\text{(bi-modal)}} = \left[\sum_{c=1}^{C_{g,ex}} \text{LN}\big(f_c^{\text{post}}(\mathbf{z}_{c,g,ex}^{\text{post}})\big), \quad \sum_{c=1}^{C_h} \text{LN}\big(f_c^{\text{post}}(\mathbf{z}_{c,h}^{\text{post}})\big)\right].$$

$$\mathbf{z}_{\text{patient}}^{\text{(tri-modal)}} = \left[\sum_{c=1}^{C_{g,ex}} \text{LN}\big(f_c^{\text{post}}(\mathbf{z}_{c,g,ex}^{\text{post}})\big), \quad \sum_{c=1}^{C_{g,mt}} \text{LN}\big(f_c^{\text{post}}(\mathbf{z}_{c,g,mt}^{\text{post}})\big), \quad \sum_{c=1}^{C_h} \text{LN}\big(f_c^{\text{post}}(\mathbf{z}_{c,h}^{\text{post}})\big)\right].$$

Unlike previous works that share a single f^{post} across all prototypes, using prototype-specific networks improves expressivity by enabling independent transformations for each prototype. The multimodal embedding $\mathbf{z}_{\text{patient}}$ is then passed into a final linear predictor

$$f_{\text{pred}}(x) = \theta^\top x,$$

to estimate the patient-level risk score, with $x = \mathbf{z}_{\text{patient}}$.

For survival prediction, we employ the Cox proportional hazards loss [50], which frames the task as learning a ranking over patient risks while naturally accommodating censored data (i.e., incomplete observation of event times). Its ranking objective requires batch training, and prototyping reduces computational cost, enabling multiple patients to be processed in each step. The Cox model assumes an exponential-linear hazard function:

$$\lambda(t \mid x) = \lambda_0(t)\, \exp(\theta^\top x),$$

where $\lambda(t \mid x)$ denotes the hazard rate at time t given patient features x, $\lambda_0(t)$ is the baseline hazard, and θ denotes the parameters of the final linear predictor f_{pred}. Under this formulation, the probability that patient i experiences the event at time t_i is:

$$P_i = \frac{\lambda(t_i \mid x_i)}{\sum_{k \in R_i} \lambda(t_i \mid x_k)} = \frac{\exp(\theta^\top x_i)}{\sum_{k \in R_i} \exp(\theta^\top x_k)},$$

where R_i denotes the set of patients uncensored and at risk at t_i. This defines a partial likelihood of each observed event , as it depends only on the relative ordering of events without requiring estimation of the baseline hazard. The negative log-partial likelihood, which is minimised during training, is:

$$l(\theta) = -\sum_{i \in U} \log \left(\frac{\exp(\theta^\top x_i)}{\sum_{k \in R_i} \exp(\theta^\top x_k)} \right) = -\sum_{i \in U} \Big(\theta^\top x_i \; - \; \log \sum_{k \in R_i} \exp(\theta^\top x_k) \Big),$$

where U is the set of uncensored patients. This objective drives the model to assign higher risk scores to patients with earlier events, thereby learning the survival ranking.

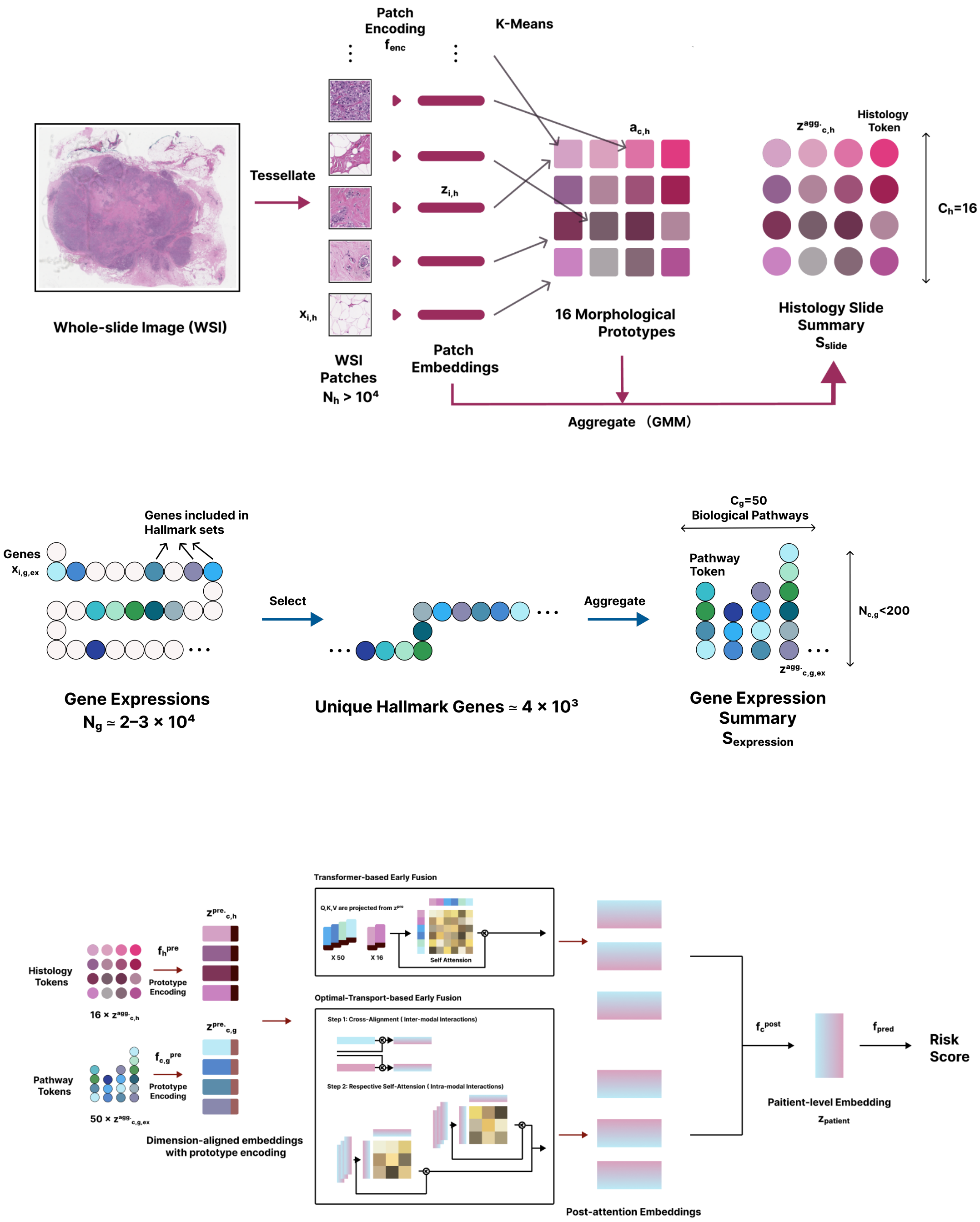

Figure 2: Overview of the bi-modal prototyping framework. (**A**) Whole slide images (WSIs) are divided into patches, which are encoded and assigned to histology prototypes representing distinct morphological patterns. (**B**) Gene expression profiles are embedded and clustered into transcriptomic prototypes capturing pathway-level variation. (**C**) The learned histology and transcriptomic embeddings are fused via either Transformer or OT cross alignment to predict patient outcomes.

3 Experiments

3.1 Datasets and Splits

All experiments were conducted on The Cancer Genome Atlas (TCGA) to evaluate MMP across four cancer types: breast invasive carcinoma (BRCA, $n = 868$), lung adenocarcinoma (LUAD, $n = 412$), stomach adenocarcinoma (STAD, $n = 318$), and kidney renal clear cell carcinoma (KIRC, $n = 340$) [51]. All reproduction and extension experiments strictly adhered to the original data partitioning, employing 5-fold site-stratified cross-validation to fully and unbiasedly utilise the limited patient samples.

Models were trained to predict patients' risk of disease-specific survival (DSS), defined as the interval from initial diagnosis to death directly caused by the diagnosed cancer, which is considered clinically relevant for assessing therapeutic impact [52]. Model performance on the held-out test folds was evaluated using the concordance index (C-Index), which measures the probability that, for any comparable pair of patients, the predicted risk scores correctly reflect the observed ordering of survival times [53]. Formally, it is defined as:

$$C = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \mathbb{1}\big(f(x_j) > f(x_i)\big),$$

where $|\mathcal{E}|$ denotes the total number of comparable pairs, and $\mathbb{1}(\cdot)$ is the indicator function. A pair (i, j) is considered comparable if patient j has a shorter observed survival time and was not censored (i.e., the disease-specific death occurred). The function $f(x)$ represents the predicted risk score, where higher values indicate greater mortality risk. A higher C-Index indicates greater predictive accuracy.

For histology data, patch embeddings were extracted using UNI, a DINOv2-based ViT-Large model pretrained on 10^8 patches sampled from 10^5 whole-slide images collected at Mass General Brigham. UNI was not pretrained on TCGA, thereby avoiding data contamination.

For gene expression data, we used log2-transformed transcripts-per-million (TPM) bulk RNA sequencing profiles sourced from the UCSC Xena database [54]. Biological pathway prototypes corresponding to 50 Hallmark Gene Sets were derived from the Molecular Signatures Database (MSigDB) [44]. These prototypes comprise 4,241 unique genes distributed across 50 pathways representing key human biological processes, with pathway sizes ranging from 31 to 199 genes.

For mutation data, we used TCGA somatic mutation profiles obtained from the Genomic Data Commons (GDC) portal [55]. Raw mutation annotation format (MAF) files were processed to extract mutated genes mapped to the same 50 Hallmark Gene Sets used to construct the gene expression prototypes. Each gene was first encoded using either binary or weighted VEP-based values, as described in Section 2.1.2, and then aggregated into biological pathway prototypes.

3.2 Reproduction Baselines

Unimodal baseline. We independently trained unimodal models on histology and gene expression data within the MMP framework, employing identical prototyping tools while excluding any modality fusion.

Multimodal baseline. We implemented SurvPath [56], a recent model that similarly encodes histology and genomic tokens with a Transformer for cancer survival prediction. SurvPath represents genomic features using 331 biological pathways and retains numerous histology patch embeddings, fusing the modalities through a memory-efficient Transformer.

MMP tests. We trained MMP models using both Transformer-based fusion ($\text{MMP}_{\text{Trans}}$) and optimal transport-based fusion (MMP_{OT}).

3.3 Extensions

Extension 1: Impact of Patch Size. We conducted an additional ablation study on the patch size used for extracting WSI histology features. Whole-slide images were tessellated into non-overlapping patches of 2048×2048 pixels at 40x magnification, with all other prototyping steps unchanged. Using the corresponding new morphology prototypes and the same biological pathway prototypes, we evaluated the impact on cancer survival prediction with Transformer-based fusion within the MMP framework.

Extension 2: Addition of DNA mutation data as a third modality. We performed tri-modal prototyping for survival prediction by incorporating mutation information with Transformer-based fusion. Additional ablation experiments examined (1) mutation prototype encodings, comparing shared encodings (the same prototype encodings used for gene expression) versus independent mutation-specific prototype encodings; and (2) mutation value encodings, comparing binary and weighted representations as described in Section 2.1.2.

3.4 Implementation

Experiments were performed on NVIDIA Ampere GPUs. Training was conducted over 20 epochs using AdamW optimization, with an initial learning rate of 1×10^{-4}, a cosine decay schedule, and a weight decay of 1×10^{-5}. For MMP, the Cox proportional hazards loss was applied with a batch size of 64. In contrast, SurvPath, serving as the non-prototype baseline, was optimized with the negative log-likelihood survival loss and trained with a batch size of 1. To improve sampling diversity and reduce GPU memory consumption during training, SurvPath randomly selected 4,096 patches per WSI. In the inference phase, predictions were made using the entire WSI without subsampling.

4 Results and Discussion

4.1 Survival Prediction

All results are presented in Table 3.

Table 3: Reproduction of baseline models, MMP methods, and subsequent extensions for disease-specific survival prediction, evaluated using the C-Index (reported as mean ± standard deviation across five folds). The clinical baseline is also provided as a reference. Grey numbers correspond to the results reported in *the original MMP study* [1]. For each cancer type, the highest performance among the reproduced methods is indicated in **bold**, while the best performance overall, including all extensions, is underlined. For $\text{MMP}_{\text{Trans}}$ with a 2048 WSI patch size, all other settings match the 1024 configuration. For the tri-modal extension, we use shared prototype encodings with gene expression and weighted mutation value encodings.

Patch Size	Dataset	BRCA	LUAD	STAD	KIRC	Avg.
	Clinical	0.563±0.055	0.528±0.028	0.592±0.044	0.602±0.066	0.571
5*1024	Histology Only	0.677±0.091	0.553±0.075	0.561±0.081	**0.753±0.177**	0.636
		(0.669±0.119)	(0.600±0.039)	(0.488±0.093)	(0.701±0.177)	(0.615)
	Gene Only	0.658±0.059	0.593±0.043	0.557±0.061	0.702±0.093	0.627
		(0.615±0.054)	(0.626±0.077)	(0.566±0.080)	(0.681±0.090)	(0.622)
	SurvPath	0.702±0.056	0.577±0.077	0.574±0.038	0.706±0.112	0.640
		(0.709±0.062)	(0.612±0.060)	(0.556±0.136)	(0.738±0.131)	(0.654)
	$\text{MMP}_{\text{Trans.}}$	**0.775±0.041**	**0.620±0.036**	0.613±0.090	0.731±0.120	**0.685**
		(0.738±0.069)	(0.642±0.037)	(0.598±0.051)	(0.747±0.106)	(0.681)
	$\text{MMP}_{\text{O.T.}}$	0.773±0.042	0.605±0.049	**0.628±0.102**	0.711±0.088	0.679
		(0.753±0.069)	(0.643±0.013)	(0.580±0.071)	(0.748±0.099)	(0.681)
Extension						
2048	$\text{MMP}_{\text{Trans.}}$ (Patch=2048)	0.713±0.094	0.633±0.041	0.587±0.067	0.723±0.102	0.664
1024	$\text{MMP}_{\text{Trans.}}$ (+ Mutation)	0.769±0.037	0.635±0.038	0.621±0.075	0.748±0.108	0.694

Reproduction Results. Both models within the MMP framework demonstrate strong predictive performance. Under the original settings reported in *the original MMP study*, the transformer-based MMP achieves the highest C-index, exceeding the best-performed unimodal approach by 7.7% and the multimodal baseline SurvPath by 7.0%. In three of the four cancer cohorts, MMP models deliver the most accurate survival predictions as shown in bold in Table 3.

The reproduced results do not exactly match those initially reported, likely due to differences in patch size and magnification during WSI embedding extraction, as discussed in Section 2.1.1, as well as variations in the high-performance computing environment. Nevertheless, the overall trends and value ranges remain consistent across models and cancer types. All subsequent analyses based on these reproduced results support the original findings. Therefore, the MMP framework proves to be a promising approach.

Comparison with Clinical Baseline. The multimodal clinical baseline incorporates age, sex, and cancer grade as defined in the TCGA cohort. As expected, all machine learning models substantially outperformed the clinical baseline, demonstrating their capability to capture complex patterns that are often difficult for clinicians to leverage directly. Among these models, both MMP variants were the only ones to consistently surpass the clinical baseline in all four cancer cohorts, which highlights their robustness and superiority.

Unimodal vs. Multimodal. On average, all three multimodal approaches outperform the prototype-enhanced unimodal baselines, which rely solely on either histology or gene expression data. This underscores the effectiveness of combining complementary information from both modalities, which are strongly associated with patient survival. Notably, for KIRC, the unimodal histology-based prediction achieved the highest accuracy among all cases. This result demonstrates the high quality of the morphological prototypes developed in this study, while also indicating that current multimodal fusion techniques have room for improvement in balancing modalities and preserving the most valuable information from each data source.

Prototypes vs. Non-Prototypes. We compare three multimodal models that all implement early fusion strategies. Both MMP models clearly outperform the multimodal baseline, SurvPath, across all cancer cohorts. SurvPath applies prototyping only to gene expression data while retaining large numbers of raw WSI patch embeddings as non-prototypes. This design increases computational burden and prevents effective modelling of intra-modality histology interactions during self-attention. It also leads to information loss when approximating attention with memory-efficient methods. Furthermore, due to the large volume of WSI patches involved in training, SurvPath cannot process multiple patients in parallel and must rely on negative log likelihood (NLL) loss, which operates on one patient at a time and is considered less flexible and less effective for survival prediction compared to the Cox loss [1]. In contrast, MMP applies high-quality prototyping to all data modalities, significantly reducing computational demands. This enables the simultaneous capture of intra- and inter-modality interactions and the use of Cox survival loss, all of which contribute to its substantial improvement in predictive performance.

Transformer vs. OT-based Cross-Attention. The performance difference between the transformer-based and OT-based MMP models is relatively small, with the transformer variant achieving slightly higher accuracy (+0.9%). This contrasts with the original MMP study, where both models were reported to achieve identical results. The discrepancy likely arises from the greater stability of parameter updates in modern transformers trained via backpropagation and enhanced by residual connections, compared to the iterative optimisation of OT-based methods with the Sinkhorn algorithm. Another advantage of the transformer-based MMP is its simpler fusion mechanism: a single attention matrix captures all interactions directly, whereas OT-based MMP employs a more complex two-step process. Therefore, the transformer-based MMP is selected for all extensions proposed in this study.

Extensions. Two extensions were conducted to further improve survival prediction accuracy.

- **Impact of WSI Patch Size.** The results in Table 3 indicate that doubling the current WSI patch size reduces model performance by 3.1%. This is primarily due to the loss of fine details. A patch size of 2048 covers four times the area of the original 1024 patch size and increases the chance of including heterogeneous morphological features. In addition, when encoded into embeddings of the same dimensionality, the effective resolution is reduced by a factor of four, which dilutes important local information. All of these make it harder for the model to learn morphological prototypes effectively under unsupervised learning. Therefore, a patch size of 1024 was selected for all subsequent experiments.

- **Incorporate Mutation Information as a Third Modality.** We explore the potential of MMP to incorporate DNA mutation as an additional data modality. To achieve optimal results, we conducted two ablation studies on the trimodal prototyping framework, summarised as follows.

    1. **Prototyping Encodings.** The trimodal model achieved better performance when

the mutation data used the same 50 pathway encodings as the gene expression data, prior to assembling the multimodal tokens for input into the transformer (Table 4). We hypothesise that sharing the encoder constrains both modalities to learn consistent representations of the same biological pathways and promote pathway-level alignment. This approach also eliminates the need to train a separate encoder for mutations, improving parameter efficiency and reducing the risk of overfitting.

2. **Mutation Value Encodings.** Applying impact-weighted encodings to mutation values achieved better performance. This suggests that incorporating both the frequency and importance of observed mutations integrates biological prior knowledge into the mutation tokens. As driver genetic mutations are key factors in cancer development, this design enables the model to focus on more meaningful signals and reduce the influence of noise.

Table 4: Ablation study assessing the impact of alternative strategies for mutation prototype encodings and mutation value encodings. For each configuration, the average C-Index (Avg.) computed across four cancer cohorts is reported.

Ablation	**Model**	**Avg.**	**Choice**
Mutation Prototype Encodings	Shared 50 pathway encodings with gene expression	0.694	✓
	Independent 50 pathway encodings	0.691	
Mutation Value Encodings	Binary mutation value	0.692	
	Impact-weighted mutation value	0.694	✓

We therefore selected a shared prototype encoder and impact-weighted encoding of mutation values for the final prediction, as shown in Table 3. The model combining all three modalities outperformed all uni- and bi-modal configurations, achieving consistently high prediction accuracy across cancer types. Compared to the best-performing transformer-based bi-modal approach, it increased the C-index by 1.3%, further demonstrating the value of incorporating diverse data modalities.

However, this improvement is relatively modest compared to the gains observed when adding a second modality. Despite the limited incremental benefit of introducing an additional modality to an already strong model, the small improvement can be attributed to two main factors. Firstly, unlike the highly complementary information provided by histology and gene expression, adding mutation data does not contribute entirely independent signals. Although mutations and RNA expression are not directly correlated, many key DNA mutations result in transcriptional dysregulation, such as disrupted transcription factor binding or altered gene expression levels. As a result, much of the information from mutations is already reflected in RNA expression.

Secondly, to ensure the generalisability of the prototypes across cancer types, we aggregated mutations into 50 hallmark biological pathways. While this approach effectively captures impactful genetic changes in core cellular functions, some important mutations are excluded from analysis. For instance, *PBRM1* is a gene involved in regulatory mechanisms rather than processes such as cell proliferation or apoptosis, and is therefore filtered out by the hallmark gene sets (Table 1). However, *PBRM1* mutations are well-established indicators of KIRC and play important prognostic roles [57]. The omission of such critical signals limits further improvements in predictive performance. This limitation could be addressed by incorporating additional prior biological knowledge, such as including known key mutated genes specific to each cancer type during prototyping, thereby fully leveraging the power of mutation data.

4.2 Statistical Analysis

Log-rank Test. To evaluate the effectiveness of MMP in risk stratification, we conducted log-rank tests comparing high-risk and low-risk cohorts defined by the 50th percentile of predicted risks from the transformer-based MMP and SurvPath models. Predicted risks from all five test folds across the four cancer types were pooled to calculate the p-values, as shown in Table 5. Using a significance threshold of 0.05, MMP predictions were significant in distinguishing high- and low-risk groups in all cancer cohorts, whereas SurvPath achieved significance in only three. These findings further confirm the robustness of MMP for survival prediction tasks.

Table 5: Risk stratification. Log-rank p-values are presented for high- and low-risk patient groups identified by SurvPath and MMP(Trans.). A p-value less than 0.05 indicates statistical significance.

	BRCA	**LUAD**	**STAD**	**KIRC**
SurvPath	1.12×10^{-2}	4.46×10^{-1}	1.39×10^{-2}	8.40×10^{-9}
$\mathbf{MMP_{Trans.}}$	1.05×10^{-6}	2.41×10^{-3}	4.80×10^{-2}	1.27×10^{-8}

Extension: Wilcoxon Signed-Rank Test. In this study, we conducted an additional analysis to assess the statistical significance of incorporating a third modality. Wilcoxon signed-rank tests were performed on the paired predictions across the five cross-validation folds [58]. As the predictions from cross-validation are not independent and the resulting p-values are only approximate, this analysis focuses on comparing the relative improvement in predictive significance when adding more data modalities. Specifically, we compared unimodal predictions versus bimodal predictions, and unimodal versus trimodal predictions. The better-performing unimodal configuration (histology only) was used as the baseline, and results are presented in Table 6. In two of the four cancer cohorts, adding a third modality resulted in a lower p-value compared to adding a second. This indicates that, relative to the unimodal baseline, the trimodal configuration achieves greater predictive differentiation than the bimodal configuration. The remaining cohorts showed no further p-value reduction. These findings align with the earlier discussion on both the strengths and limitations of the current approach to prototyping DNA mutation information for cancer survival prediction within the MMP framework (Section 4.1).

Table 6: Wilcoxon Signed-Rank Test comparing prediction performance. For each cancer type, we tested whether including a second (bi-modal) or third (tri-modal) modality improved performance over the best unimodal prediction (histology-only). The tests used 5-fold cross-validation pairs; since folds are not independent, p-values are approximate and reported only for relative comparison. Boldface p-values indicate cases where adding a third modality further improves statistical significance.

	BRCA	**LUAD**	**STAD**	**KIRC**
Unimodal vs Bi-modal	0.0938	0.0312	0.1562	0.6875
Unimodal vs Tri-modal	**0.0625**	0.0312	**0.0625**	0.6875

4.3 Interpretability

Unimodal (histology). We visualised the 16 unsupervised morphological prototypes learned for WSI, as illustrated in Figure 3 for a BRCA example. The original WSI image is provided as a reference. A heatmap of a randomly selected prototype shows its similarity to each patch, based on the posterior probability $p(c_i = c \mid z_i, h)$ modelled by the GMM. Most probabilities are either 0 or 1, indicating that the GMM effectively learned distinct histological features.

We also present a prototype assignment map, which labels each patch with its most representative prototype (i.e., the one with the highest posterior probability). For comparison, Figure 3(d) shows the corresponding visualisation from *the original MMP study* for the same WSI and fold. The assignments generally align, though the original study exhibits more detailed distinctions due to its smaller patch area.

Finally, we examined the probability distributions of the morphological prototypes and display example patches assigned to four prototypes. The clear and distinct patterns confirm the effectiveness of the histology prototyping approach.

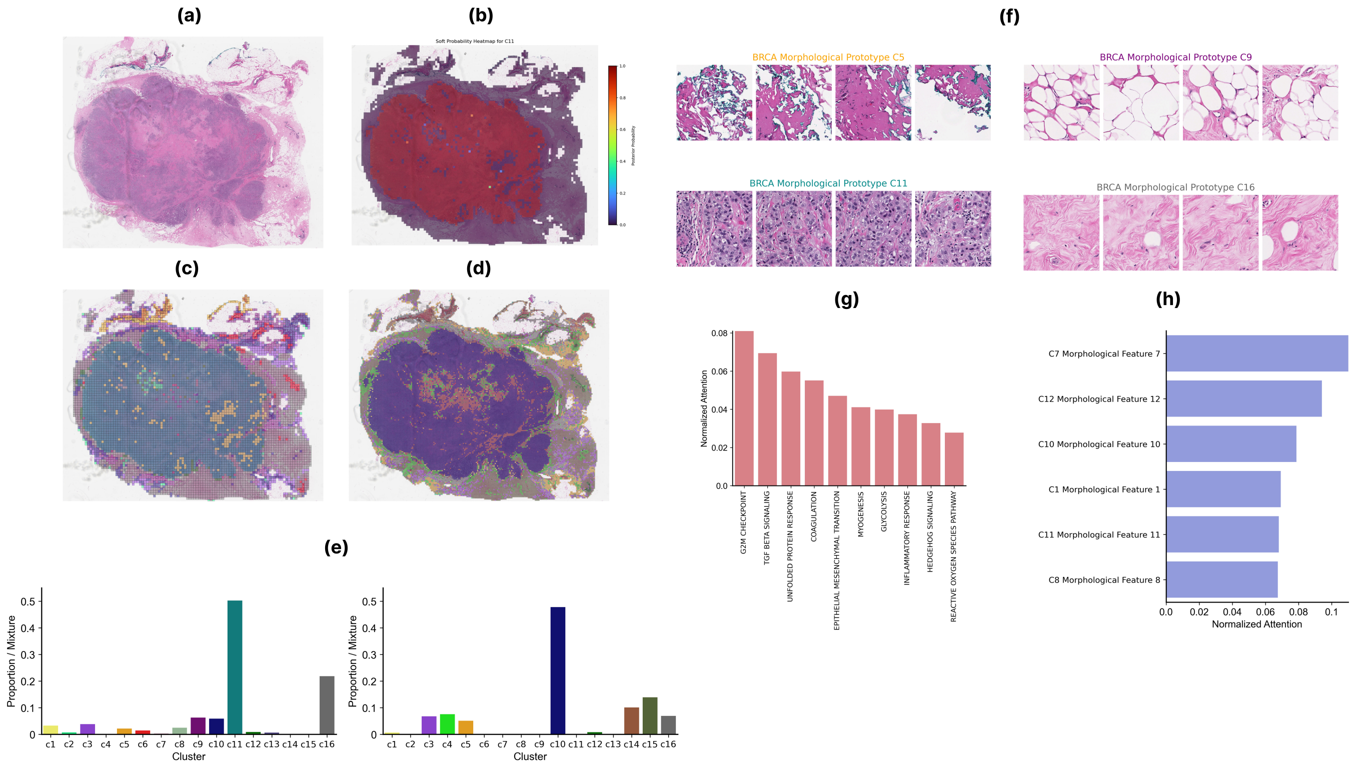

Figure 3: Visualisation of morphological prototype learning and interpretability in a BRCA sample. (a) Original WSI reference image. (b) Heatmap of posterior probabilities for C12, illustrating similarity across patches. (c) Prototype assignment map generated in this study. (d) Prototype assignment map from *the original MMP study* for comparison [1]. (e) Proportion distribution of prototypes learned in this study and in *the original MMP study*. (f) Example patches representing four morphological prototypes (C5, C9, C11, C16), demonstrating distinct histological patterns. (g) Top 10 attention weights of biological pathways attending to prototype C11. (h) Top 6 attention weights of morphological prototypes attending to the pathway *epithelial_mesenchymal_transition*.

Bimodal (histology + gene expression). MMP enables visualisation of bidirectional relationships between modalities by interpreting cross-modal attention scores. Importantly, since all prototypes are designed to have clear biological meaning, the concept-based interactions can yield valuable insights. Figure 3(g)(h) illustrates these relationships.

For a selected histology prototype, it is possible to identify the most correlated gene expression pathways. In clinical practice, if a pathologist can identify the tissue type associated with each morphological prototype, this approach provides critical information about RNA expression patterns relevant to cancer, particularly when analysing prototypes corresponding to dense tumour regions. This has great potential for uncovering hidden genetic mechanisms of cancer and informing the development of targeted therapies.

Conversely, for a chosen gene expression pathway, one can visualise the most strongly attended morphological prototypes. This perspective helps to characterise how gene expression alterations manifest at the tissue level in cancer patients.

EXTENSION: Trimodal (histology + gene expression + mutation). We implemented analogous interpretability analyses for the trimodal prototyping framework using transformer-based models. Figure 4 displays the bidirectional attention weights capturing interactions between morphological prototypes and biological pathway prototypes derived from RNA expression and DNA mutations.

In this setting, for a given morphological prototype, the top ten associated pathways identified from gene expression and mutation data exhibited a 60% overlap, with the correlations appearing in a different order. When selecting a single pathway, the respective top six most-attended morphological prototypes derived from gene expression and mutation information showed a 66% overlap.

These observations illustrate both the related and complementary roles of gene expression and genetic mutations in cancer development, highlighting opportunities for further exploration by clinical researchers.

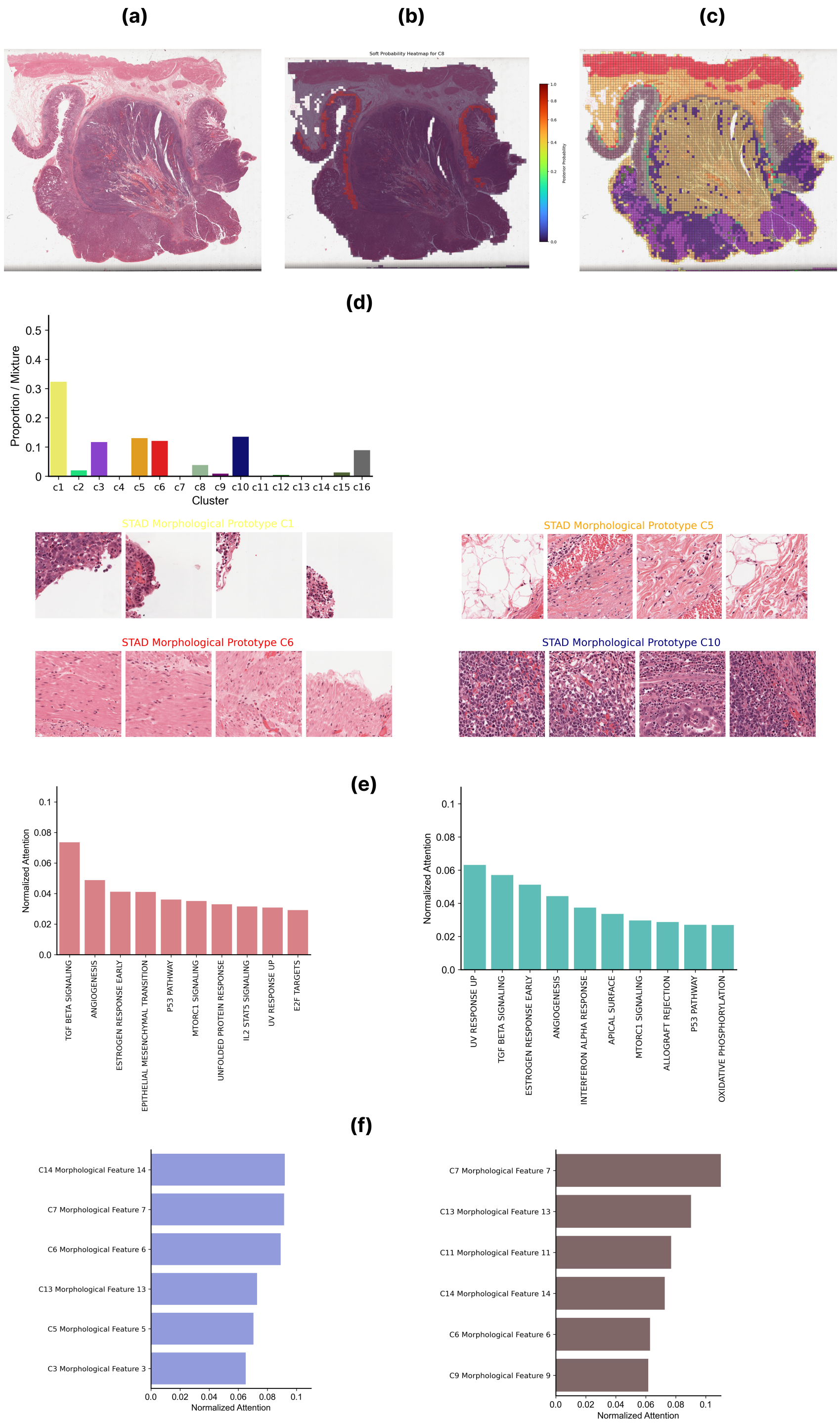

Figure 4: Visualisation of trimodal morphological prototype learning and interpretability in a STAD sample. (a) Original WSI reference image. (b) Heatmap of posterior probabilities for a randomly selected prototype. (c) Prototype assignment map indicating spatial distribution of the learned morphological clusters. (d) Top: proportion distribution of the 16 morphological prototypes identified in this sample. Bottom: representative patches for four selected morphological prototypes (C1, C3, C5, C10). (e) Top 10 attention weights of biological pathways attending to prototype C9, derived from RNA expression (left) and DNA mutation data (right). (f) Top 6 morphological prototypes attending to the pathway *tgf_beta_signaling*, based on RNA expression (left) and mutation data (right).

5 Conclusion

In this study, we reproduced the multimodal prototyping (MMP) framework for modern machine learning applications in survival prediction. Following *the original MMP study*, we first fused two data modalities for outcome prediction. To reduce computational cost and enable meaningful bidirectional interpretability, we aggregated over 10^4 WSI patch embeddings into 16 morphological prototypes and over 2×10^4 gene expressions into 50 biological pathway prototypes, obtaining corresponding modality tokens. These tokens were then combined using both transformer-based and optimal transport (OT)-based fusion strategies. Both methods significantly outperformed unimodal and multimodal baselines, with transformer-based fusion achieving the highest average C-index of 0.685 across four TCGA cancer cohorts (BRCA, LUAD, STAD, KIRC), demonstrating the strong predictive capability of MMP.

To further improve prediction accuracy, we extended the original framework in two directions. First, we showed that increasing the WSI patch size from 1024 to 2048 at 40× magnification diluted key local patterns and degraded performance. Second, we incorporated a third modality, DNA mutation data, for trimodal survival prediction with transformer-based fusion. We employed the same 50 biological hallmark pathways for pathway prototyping and used shared prototype encodings with gene expression along with impact-weighted mutation values, which yielded the best results. The final optimal model achieved an average C-index of 0.694 across the four cancer cohorts, outperforming all bimodal configurations. The moderate improvement highlights both the potential of integrating additional complementary modalities in the MMP framework and the need to further optimise the prototyping of mutation information.

Finally, we interpreted the cross-modal interactions using attention weights from transformer-based fusion in both bimodal and trimodal models. The concept-based prototypes provide valuable insights into the relationships between morphological and genomic features and can support cancer researchers in uncovering hidden patterns that may inform the development of more targeted therapies.

Future work could include: (1) exploring more cancer-specific strategies for incorporating mutation data to better exploit the genomic information driving cancer progression, and (2) extending the MMP framework to broader prognostic applications, such as recurrence risk assessment or clinically relevant tasks like predicting homologous recombination deficiency (HRD), which can inform treatment decisions and drug response in cancer patients.

6 Declaration

During the development of this project, I made use of Github Copilot's autocompletion feature to assist in coding and generating docstrings for functions. Additionally, ChatGPT was used to support debugging. For the final report, ChatGPT helped in generating LaTeX code in desired formats.

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